COSTVISION · SOFTWARE SHOULD-COST REPORT

# Range Rover L460 (PHEV)

Whole-vehicle software · PHEV · 52 software modules

China engineering hub · Tier-1 supplier · 200,000 vehicles a year × 6 years

Prepared 14 October 2026 · rate book CostVision built-in v2.0.0 (country book 2026-10) (reviewed 2026-10)

| PROGRAMME TOTAL | PER VEHICLE, ALL-IN | ENGINEERING (NRE) | LIFECYCLE |
|---|---|---|---|
| **£265.3M** | **£221** | **£46.3M** | **£219.0M** |
| ~ P16 of the simulated range | over 1,200,000 vehicles | 4,806 person-months | to programme year 16 |

## Read this first

- The figures are a deterministic should-cost model: every number comes from the module catalogue, the hub rate book and stated constants — no AI sets a price. The full arithmetic is in section 3 and was reconciled independently to the penny.
- 78.7 % of the total (£208.7M) rests on unsourced catalogue figures — chiefly cloud, map data, the OS licence placeholder and the module effort catalogue. Treat them as estimates to replace with supplier quotes.
- Accuracy against real programme costs has not been measured: no actual programme cost has been logged in the tool.
- The vehicle facts behind the scope were checked against the manufacturer's and press pages (section 2). The annual volume and programme life are the user's inputs, not published figures.

| | |
|---|---|
| **Programme** | Range Rover L460 (PHEV) |
| **Software scope** | 52 modules in 7 domains (section 5); excluded modules and catalogue gaps in section 12 |
| **Engineering hub** | China — plus every hub in the rate book re-costed (section 8) |
| **Countries** | 39 manufacturing countries mapped to their software hub (section 8) |
| **Currency** | £ — engine prices in £; figures shown in the report currency |

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# 1 Executive summary

Range Rover L460 (PHEV): 52 software modules, 200,000 vehicles a year for 6 years (1,200,000 vehicles), engineered at the China hub by a tier-1 supplier.

| Figure | Value | What it covers |
|---|---|---|
| **Programme total** | **£265.3M** | all engineering and every lifecycle cost to programme year 16 |
| Per vehicle, all-in | £221 | total ÷ 1,200,000 vehicles built |
| Engineering (NRE) | £46.3M | 17.4 % — development, verification, integration, cyber, calibration, tools, homologation |
| Lifecycle | £219.0M | 82.6 % — maintenance, licences and royalties, cloud |
| Royalty per vehicle built | £37.84 | paid on every vehicle in the piece price |
| Running cost per vehicle-year | £22.25 | map data (3 yr) + cloud (8 yr) per vehicle in service |
| Simulated range | £254.3M – £358.0M | P10–P90; P50 £301.3M; the headline sits at about P16 |
| Hub range (same scope) | £227.3M – £376.4M | Egypt to USA — California (section 8) |

## What the figures say

- Lifecycle is 82.6 % of the total: cloud £103.2M, licences and royalties £97.4M, maintenance £18.4M. The running cost, more than the engineering, sets this programme's price.
- The largest domain is Infotainment, Connectivity & UX at £110.1M (41.5 %); the three largest modules — Navigation Stack £50.2M, IVI Operating System (Android Automotive / QNX) £28.8M, Cloud Backend Services £28.4M — are 40.5 % of the total.
- Engineering is 4,806 person-months at £5,573 per person-month loaded; moving the hub moves only the engineering and its maintenance — cloud, royalties and licences do not change with the hub.
- The total rests mostly on unsourced catalogue figures (78.7 %); the published-source share is 8.6 %. Replacing the largest unsourced lines with supplier quotes (section 10) is the fastest way to firm up the figure.

# 2 Programme definition

Every input the costing used. Defaults are the tool's stated assumptions; the basis of each is in section 10.

| Input | Value | Basis |
|---|---|---|
| Vehicle / programme | Range Rover L460 (PHEV) | vehicle demo (facts below) |
| Drivetrain | PHEV | engine powertrain scope |
| Annual volume | 200,000 | user input |
| Programme life (production years) | 6 | user input |
| Engineering hub | China | hub rate book (appendix) |
| Development source | Tier-1 supplier | modelling assumption; efficiency factor unsourced, supplier margin sourced (section 10) |
| Senior engineer share | 55 % | demo / user input |
| Engineering burden | employment +15 %, company +40 %, capital +0 % | section 10 |
| Reuse (all modules unless set) | Medium | tool default Medium (× 0.60) — unsourced |
| Development window | 54 months | Volkswagen's stated cycle (section 10) |
| Connected service per vehicle | 8 years | median of current OEM terms |
| Map-data subscription per vehicle | 3 years | OEM map-update terms |
| Security support after the last build | 10 years | UN R155 post-production duty; share assumed |
| Discount rate | 0 % (nominal money) | user input |
| Homologation | included | section 10 |

## The vehicle

Range Rover (L460, 2022 on): JLR MLA-Flex platform, EVA 2.0 electrical architecture, Pivi Pro infotainment with software-over-the-air updates. PHEV: P460e / P550e, 38.2 kWh battery (MY24 on). Premium software from JLR-listed features: Dynamic Response Pro (48 V active roll control), all-wheel steering, Meridian Signature Sound, Cabin Air Purification Pro, head-up display (option), 3D surround camera and Remote Park Assist. Voice: Amazon Alexa. Digital key is not confirmed for the L460 and is out of scope. Tier-1 development is a modelling assumption, not JLR's sourcing.

## Product facts the scope rests on (each checked on the page given)

| Fact | Checked on |
|---|---|
| MLA-Flex platform; EVA 2.0 electrical architecture; Pivi Pro with software-over-the-air updates | https://media.landrover.com/introducing-new-range-rover-seamlessly-connected-convenience-and-luxury-every-turn |
| PHEV P460e / P550e, 38.2 kWh battery, up to 75 miles WLTP (from MY24) | https://www.fleetnews.co.uk/news/manufacturer-news/2023/05/17/range-rover-plug-in-hybrids-updated-with-more-power-and-range |
| Dynamic Response Pro (48 V electronic active roll control); all-wheel steering | https://media.landrover.com/introducing-new-range-rover-serene-capability-assurance-and-composure-and-road |
| Meridian Signature Sound System (35 loudspeakers) | https://meridian-audio.com/media/news-events/meridian-audio-brings-best-ever-audio-system-for-new-range-rover-to-mark-10-years-of-partnership/ |
| Cabin Air Purification Pro (PM2.5 filtration) | https://www.landrover.in/range-rover/range-rover/explore.html |
| Head-up display (option), 3D surround camera, Remote Park Assist | https://media.landrover.com/introducing-new-range-rover-seamlessly-connected-convenience-and-luxury-every-turn |
| Amazon Alexa is the Pivi Pro voice assistant | https://www.landroverusa.com/our-story/news/amazon-alexa.html |
| No published software-cost figure for the L460 was found; no model-level volume is published (JLR reports Range Rover family wholesales) | https://media.jaguarlandrover.com/news/2024/04/jlr-full-year-sales-over-20-cent-year-year-reflecting-improved-production-and |

# 3 How the cost is built

The seven steps the tool runs, with this programme's figures. An independent script recomputed every module and bucket from the same published tables and matched the tool to the penny at every hub (scripts/sw-review/rr-phev-trace.ts).

| # | Step | Result | How |
|---|---|---|---|
| 1 | **Scope** | 52 modules | 7 software domains on a PHEV vehicle; nominal effort 2,472 person-months (fresh code, QM, medium complexity). |
| 2 | **Development effort** | 2,779 PM | Reuse, complexity and the ISO 26262 ASIL uplift applied to each module: 2,472 -> 2,779 PM. 18 modules at ASIL C/D (lit). |
| 3 | **Verification & integration** | +2,027 PM | SIL / MIL / HIL testing, integration, ISO/SAE 21434 cybersecurity and calibration: 4,806 PM of engineering in all. 20 modules at CAL3/4 (lit). |
| 4 | **Engineering rate** | £5,573 / PM | China: 55 % senior blend of the hub's mid and senior rates; mid £2,828 / senior £4,001 per person-month (salary + statutory on-cost ÷ productive hours × 152 h), × development source price × 1.61 engineering burden. |
| 5 | **Engineering cost (NRE)** | £46.3M | Effort × rate = £26.8M; tools £15.8M, homologation £3.7M. Largest domain: Infotainment, Connectivity & UX £110.1M (all costs). |
| 6 | **Lifecycle** | £219.0M | Maintenance £18.4M, cloud £103.2M, licences and royalties £97.4M — paid over the 6 production years and the service years after the last build (to programme year 16). |
| 7 | **Per vehicle** | £221 | NRE £46.3M over 1,200,000 vehicles (6-year recovery) + lifecycle over 1,200,000 vehicles. Programme total £265.3M. |

## Formulas

- Loaded rate = hub mid / senior blend × development-source price factor × engineering burden.
- Development person-months = catalogue person-months × reuse × ASIL factor; costed across requirements 12 %, architecture 14 %, algorithms 22 % × complexity, implementation 37 % × (1 + 0.15 × (complexity - 1)), safety 15 % (reuse floored by ASIL).
- Testing, integration, cybersecurity (by the ISO/SAE 21434 CAL) and calibration are fractions of development; tools = engineer-years × the seat price.
- Maintenance = (development + testing + integration) × the module's % a year — full for 3 years after SOP, then 50 %, then the post-production security share after the last build.
- Royalties = vehicles × £ per vehicle built + vehicle-years × £ per vehicle-year; cloud = connected vehicle-years × £ per connected vehicle-year.

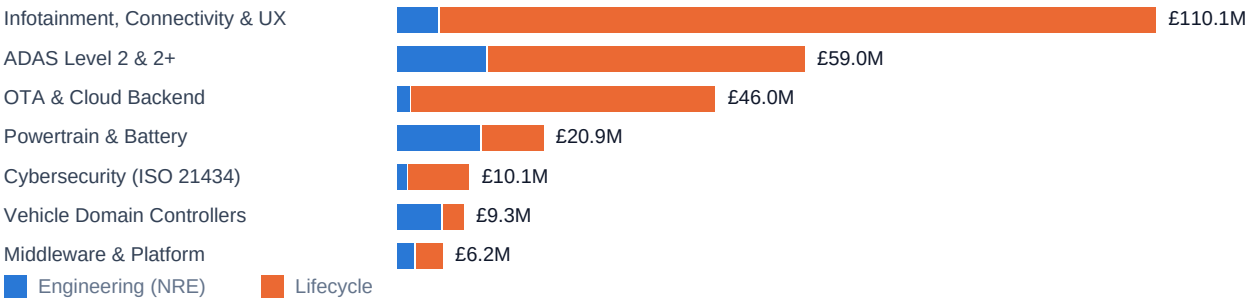
# 4 Cost build-up

Every bucket, the two subtotals and the total. The check line confirms the buckets add up.

| Bucket | £M | Share |
|---|---|---|
| Development engineering | £15.5M | 5.8 % |
| Testing and validation (SIL / MIL / HIL) | £6.4M | 2.4 % |
| Integration | £2.9M | 1.1 % |
| Cybersecurity engineering | £0.8M | 0.3 % |
| Calibration | £1.3M | 0.5 % |
| Development tools | £15.8M | 6.0 % |
| Homologation (R155 / R156 / ISO 26262) | £3.7M | 1.4 % |
| **Engineering (NRE) subtotal** | **£46.3M** | **17.4 %** |
| Maintenance and security support | £18.4M | 6.9 % |
| Licences and royalties | £97.4M | 36.7 % |
| Cloud | £103.2M | 38.9 % |
| **Lifecycle subtotal** | **£219.0M** | **82.6 %** |
| **Programme total** | **£265.3M** | **100.0 %** |

Check, exact: NRE £46,256,466 + lifecycle £219,023,715 = £265,280,182; the programme total is £265,280,182 (they agree).

## By software domain




| Domain | Modules | NRE | Lifecycle | Total | Share |
|---|---|---|---|---|---|
| Infotainment, Connectivity & UX | 8 | £5.9M | £104.2M | £110.1M | 41.5 % |
| ADAS Level 2 & 2+ | 9 | £12.8M | £46.1M | £59.0M | 22.2 % |
| OTA & Cloud Backend | 4 | £1.8M | £44.2M | £46.0M | 17.3 % |
| Powertrain & Battery | 13 | £12.0M | £8.9M | £20.9M | 7.9 % |
| Cybersecurity (ISO 21434) | 5 | £1.4M | £8.8M | £10.1M | 3.8 % |
| Vehicle Domain Controllers | 7 | £6.3M | £2.9M | £9.3M | 3.5 % |
| Middleware & Platform | 6 | £2.4M | £3.9M | £6.2M | 2.4 % |

Domain totals are module costs; homologation (programme-level) sits outside the domains.

# 5 Every software module

All 52 modules in scope, grouped by domain. PM = costed engineering person-months (development, verification, integration, cyber, calibration). NRE includes the module's tools; lifecycle is its maintenance, licences, royalties and cloud.

| Module | ASIL | Complexity | Reuse | CAL | PM | NRE | Lifecycle | Total |
|---|---|---|---|---|---|---|---|---|
| **Infotainment, Connectivity & UX** | | | | | | | | |
| Navigation Stack | QM | High | Medium | — | 44 | £0.4M | £49.9M | £50.2M |
| IVI Operating System (Android Automotive / QNX) | QM | Very High | Medium | CAL4 | 248 | £2.2M | £26.6M | £28.8M |
| Telematics Control Unit (TCU) Software | B | High | Medium | CAL4 | 51 | £0.5M | £20.4M | £20.8M |
| Vehicle Voice Assistant | QM | Very High | Medium | CAL4 | 100 | £0.9M | £5.1M | £5.9M |
| App Framework & HMI Layer | QM | Very High | Medium | — | 112 | £1.0M | £1.0M | £2.0M |
| Bluetooth / WiFi / 5G Connectivity Stack | QM | High | Medium | CAL4 | 34 | £0.3M | £0.7M | £1.0M |
| Premium Audio DSP & Active Noise Control | QM | Very High | Medium | — | 38 | £0.3M | £0.3M | £0.7M |
| AR Head-Up Display & AR Navigation | A | High | Medium | — | 40 | £0.4M | £0.2M | £0.6M |
| **ADAS Level 2 & 2+** | | | | | | | | |
| Camera Perception Stack | B | Very High | Medium | CAL3 | 304 | £2.7M | £20.9M | £23.6M |
| Sensor Fusion Engine | C | Very High | Medium | — | 246 | £2.2M | £5.9M | £8.1M |
| Highway Assist / Traffic Jam Assist | C | Very High | Medium | — | 196 | £1.7M | £6.1M | £7.9M |
| Path Planning Algorithms | C | Very High | Medium | — | 202 | £1.8M | £4.6M | £6.4M |
| Driver Monitoring Software (DMS) | B | Very High | Medium | CAL2 | 121 | £1.1M | £5.0M | £6.0M |
| Radar Processing Stack | B | Very High | Medium | — | 127 | £1.1M | £2.5M | £3.6M |
| Control Algorithms (ACC / LKA / AEB) | D | High | Medium | — | 152 | £1.4M | £0.8M | £2.1M |
| Automated Parking & Surround-View | B | High | Medium | — | 81 | £0.7M | £0.3M | £1.0M |
| Ultrasonic Processing Software | B | Medium | Medium | — | 19 | £0.2M | £0.1M | £0.3M |
| **OTA & Cloud Backend** | | | | | | | | |
| Cloud Backend Services | QM | Very High | Medium | CAL4 | 96 | £0.9M | £27.5M | £28.4M |
| Data Pipeline & Telemetry | QM | High | Medium | CAL3 | 34 | £0.3M | £7.5M | £7.8M |
| OTA Update Manager (Vehicle-side) | B | High | Medium | CAL4 | 52 | £0.5M | £4.4M | £4.9M |
| Fleet Management & Analytics Software | QM | Medium | Medium | — | 17 | £0.2M | £4.7M | £4.9M |
| **Powertrain & Battery** | | | | | | | | |
| SOC/SOH/SOE Estimation Models | C | High | Medium | — | 81 | £0.7M | £3.1M | £3.8M |
| Hybrid Supervisory Control & Energy Management | C | Very High | Medium | — | 210 | £1.9M | £0.9M | £2.8M |
| BMS Core Software | D | High | Medium | CAL4 | 211 | £1.9M | £0.8M | £2.7M |
| EDU (Electric Drive Unit) Control Software | D | High | Medium | CAL4 | 159 | £1.4M | £0.7M | £2.1M |
| Engine Management Software (ECU) | B | Very High | Medium | CAL3 | 150 | £1.3M | £0.6M | £2.0M |
| Inverter Control Algorithms | D | Very High | Medium | — | 125 | £1.1M | £0.5M | £1.6M |
| Motor Control (FOC/DTC/SVPWM) | D | Very High | Medium | — | 118 | £1.0M | £0.4M | £1.5M |
| Transmission Control Software (TCU) | B | High | Medium | — | 103 | £0.9M | £0.4M | £1.3M |
| Fast-Charging Control Software | C | Medium | Medium | CAL3 | 41 | £0.4M | £0.9M | £1.2M |
| Battery Thermal Management Software | B | High | Medium | — | 48 | £0.4M | £0.2M | £0.6M |
| Regenerative Braking Software | C | High | Medium | — | 38 | £0.3M | £0.1M | £0.5M |
| Cell Balancing Algorithms | C | High | Medium | — | 37 | £0.3M | £0.1M | £0.5M |
| Emissions After-treatment & OBD | QM | High | Medium | — | 34 | £0.3M | £0.1M | £0.4M |
| **Cybersecurity (ISO 21434)** | | | | | | | | |
| Secure OTA Software Update Framework | B | High | Medium | CAL4 | 35 | £0.3M | £3.0M | £3.3M |
| Intrusion Detection System (IDS) | QM | High | Medium | CAL3 | 35 | £0.3M | £2.4M | £2.7M |

| Module | ASIL | Complexity | Reuse | CAL | PM | NRE | Lifecycle | Total |
|---|---|---|---|---|---|---|---|---|
| Key Management System | B | High | Medium | CAL4 | 26 | £0.2M | £2.0M | £2.2M |
| Secure Boot & Root of Trust | B | High | Medium | CAL3 | 36 | £0.3M | £1.2M | £1.6M |
| Encryption Modules & Crypto Stack | B | Medium | Medium | CAL3 | 21 | £0.2M | £0.2M | £0.3M |
| **Vehicle Domain Controllers** | | | | | | | | |
| Vehicle Motion Management (VMM) | C | Very High | Medium | — | 210 | £1.9M | £0.9M | £2.8M |
| Zonal Architecture Software | B | Very High | Medium | CAL4 | 145 | £1.3M | £0.6M | £1.9M |
| Chassis Control Software | C | High | Medium | — | 118 | £1.0M | £0.5M | £1.5M |
| Predictive Active Suspension & Body Control | C | Very High | Medium | — | 109 | £1.0M | £0.4M | £1.4M |
| Body Control Software | B | Medium | Medium | — | 58 | £0.5M | £0.2M | £0.7M |
| Gateway ECU Software | B | High | Medium | CAL4 | 47 | £0.4M | £0.2M | £0.6M |
| Climate & Thermal Comfort Control | A | Medium | Medium | — | 28 | £0.2M | £0.1M | £0.3M |
| **Middleware & Platform** | | | | | | | | |
| AUTOSAR Adaptive Platform | B | High | Medium | CAL3 | 105 | £0.9M | £1.7M | £2.6M |
| AUTOSAR Classic BSW & Integration | C | High | Medium | — | 82 | £0.7M | £1.0M | £1.7M |
| Diagnostics (UDS / OBD-II) | A | Medium | Medium | — | 26 | £0.2M | £0.7M | £0.9M |
| CAN / LIN / FlexRay / Ethernet Communication Stacks | B | Medium | Medium | — | 25 | £0.2M | £0.2M | £0.4M |
| RTOS & OS Porting | C | Medium | Medium | — | 18 | £0.2M | £0.2M | £0.4M |
| Time Synchronisation & Network Management | A | Low | Medium | — | 13 | £0.1M | £0.1M | £0.2M |
| **All modules** | | | | | **4,806** | **£42.6M** | **£219.0M** | **£261.6M** |

Check, exact: module sum £261,624,781 + homologation £3,655,401 = programme total £265,280,182.

# 6 Lifecycle and unit economics

The figures purchasing compares, and every per-vehicle charge with the basis it was priced on.

| Figure | Value | Volume | Programme £ |
|---|---|---|---|
| Engineering, one-off (NRE) | £46.3M | — | £46.3M |
| Royalty per vehicle built | £37.84 | 1,200,000 vehicles | £45.4M |
| Map data per vehicle-year | £11.50 | 3,600,000 vehicle-years (3 yr each) | £41.4M |
| Cloud per connected vehicle-year | £10.75 | 9,600,000 vehicle-years (8 yr each) | £103.2M |
| Fixed per production year (maintenance, IP licences, platform) — average | £4.8M | × 6 years | £29.0M |
| **Programme total** | | | **£265.3M** |

## Royalties per vehicle — each priced on its own basis

| Module | Programme £ | Grade | Basis |
|---|---|---|---|
| Navigation Stack | £41.4M | Unsourced catalogue | Map data £8–15 / vehicle / yr — midpoint £11.50, an unsourced estimate: neither HERE nor TomTom publishes a per-car price (TomTom FY25: automotive revenue €322M, backlog €2.4bn, no vehicle count), paid for the subscription years after each vehicle is built (default 3 — OEM map-update terms). |
| IVI Operating System (Android Automotive / QNX) | £22.7M | Unsourced catalogue | Unsourced placeholder ~$25 / vehicle for Google Automotive Services (GAS) or a QNX runtime licence — neither price is public: Google does not publish a GAS fee, and BlackBerry licenses QNX per unit but publishes no per-vehicle royalty (FY26 QNX revenue US$268M, royalty backlog ~US$950M, 275M+ vehicles — SEC 10-K). Android Automotive OS itself is open source with no licence fee. Replace with your supplier quote. |
| Telematics Control Unit (TCU) Software | £18.1M | Published source | Avanci 4G vehicle licence (cellular standard-essential patents) US$20 per vehicle, paid once by the car maker — avanci.com published rate (5G US$32, 3G US$9, eCall only US$3), at the app's USD_PER_GBP. Set to 0 if your TCU supplier's price already carries it. |
| Vehicle Voice Assistant | £4.6M | Published source | Cerence average price per unit US$5.09 (TTM to Q2 FY2026; US$5.05 at Q4 FY2025; includes connected services) at the app's USD_PER_GBP — Cerence earnings decks filed with the SEC (EX-99), read via search results. |

## Embedded IP licences (£ a year per module, over the production years)

| Module | Programme £ | Module | Programme £ |
|---|---|---|---|
| Camera Perception Stack | £1.0M | AUTOSAR Classic BSW & Integration | £0.8M |
| Cloud Backend Services | £0.8M | AUTOSAR Adaptive Platform | £0.7M |
| Driver Monitoring Software (DMS) | £0.5M | App Framework & HMI Layer | £0.5M |
| Highway Assist / Traffic Jam Assist | £0.4M | Data Pipeline & Telemetry | £0.4M |
| Sensor Fusion Engine | £0.3M | OTA Update Manager (Vehicle-side) | £0.3M |
| Path Planning Algorithms | £0.3M | Intrusion Detection System (IDS) | £0.3M |
| Fleet Management & Analytics Software | £0.3M | Premium Audio DSP & Active Noise Control | £0.3M |
| Radar Processing Stack | £0.2M | Bluetooth / WiFi / 5G Connectivity Stack | £0.2M |
| RTOS & OS Porting | £0.2M | Secure OTA Software Update Framework | £0.2M |
| Vehicle Motion Management (VMM) | £0.2M | Key Management System | £0.2M |
| Hybrid Supervisory Control & Energy Management | £0.2M | SOC/SOH/SOE Estimation Models | £0.1M |
| EDU (Electric Drive Unit) Control Software | £0.1M | Control Algorithms (ACC / LKA / AEB) | £0.1M |
| Zonal Architecture Software | £0.1M | CAN / LIN / FlexRay / Ethernet Communication Stacks | £0.1M |
| Secure Boot & Root of Trust | £0.1M | Engine Management Software (ECU) | £0.1M |
| BMS Core Software | £0.1M | Telematics Control Unit (TCU) Software | £0.1M |
| Chassis Control Software | £0.1M | AR Head-Up Display & AR Navigation | £0.1M |
| Transmission Control Software (TCU) | £0.1M | Diagnostics (UDS / OBD-II) | £0.1M |

| Module | Programme £ | Module | Programme £ |
|---|---|---|---|
| Encryption Modules & Crypto Stack | £0.1M | Predictive Active Suspension & Body Control | £0.1M |
| Fast-Charging Control Software | £0.1M | Inverter Control Algorithms | £0.1M |
| Motor Control (FOC/DTC/SVPWM) | £0.1M | Gateway ECU Software | £0.1M |
| Automated Parking & Surround-View | £0.1M | Body Control Software | £0.1M |
| Ultrasonic Processing Software | £0.0M | Cell Balancing Algorithms | £0.0M |
| Battery Thermal Management Software | £0.0M | Regenerative Braking Software | £0.0M |
| Time Synchronisation & Network Management | £0.0M | Climate & Thermal Comfort Control | £0.0M |
| Emissions After-treatment & OBD | £0.0M | | |

IP licences £10.6M in all — the module catalogue's £ a year, unsourced (no licence prices are published for these stacks).

## Cloud per module

| Module | Programme £ | Module | Programme £ |
|---|---|---|---|
| Cloud Backend Services | £26.2M | Camera Perception Stack | £18.5M |
| Navigation Stack | £8.3M | Data Pipeline & Telemetry | £7.0M |
| Highway Assist / Traffic Jam Assist | £4.8M | Sensor Fusion Engine | £4.4M |
| Fleet Management & Analytics Software | £4.4M | Driver Monitoring Software (DMS) | £3.9M |
| OTA Update Manager (Vehicle-side) | £3.9M | Path Planning Algorithms | £3.3M |
| SOC/SOH/SOE Estimation Models | £2.6M | IVI Operating System (Android Automotive / QNX) | £2.6M |
| Secure OTA Software Update Framework | £2.6M | Telematics Control Unit (TCU) Software | £2.0M |
| Intrusion Detection System (IDS) | £2.0M | Radar Processing Stack | £1.7M |
| Key Management System | £1.7M | Secure Boot & Root of Trust | £1.0M |
| Fast-Charging Control Software | £0.7M | AUTOSAR Adaptive Platform | £0.7M |
| Diagnostics (UDS / OBD-II) | £0.5M | Bluetooth / WiFi / 5G Connectivity Stack | £0.4M |

Cloud £103.2M in all. Basis (every module): Module cloud £ / yr ÷ reference fleet — an unsourced estimate (~ £10.8 per connected vehicle-year with every module in scope). No OEM publishes a back-end cost; the only list price, AWS IoT FleetWise (US$0.60 / 0.45 per vehicle-month, messages and storage extra; closed to new customers 30 Apr 2026), is US$5.4–7.2 per vehicle-year for data collection alone. Each connected vehicle served 8 years from build — the median of current OEM terms: GM OnStar Basics 8 yr (MY2025+), Tesla Standard Connectivity 8 yr (ordered after 20 Jul 2022), Toyota up to 10 yr, Mercedes-Benz mbrace Connect 5 yr (US); OEM pages, re-checked 14 Oct 2026. A 2G / 3G / 4G shutdown can end it sooner.

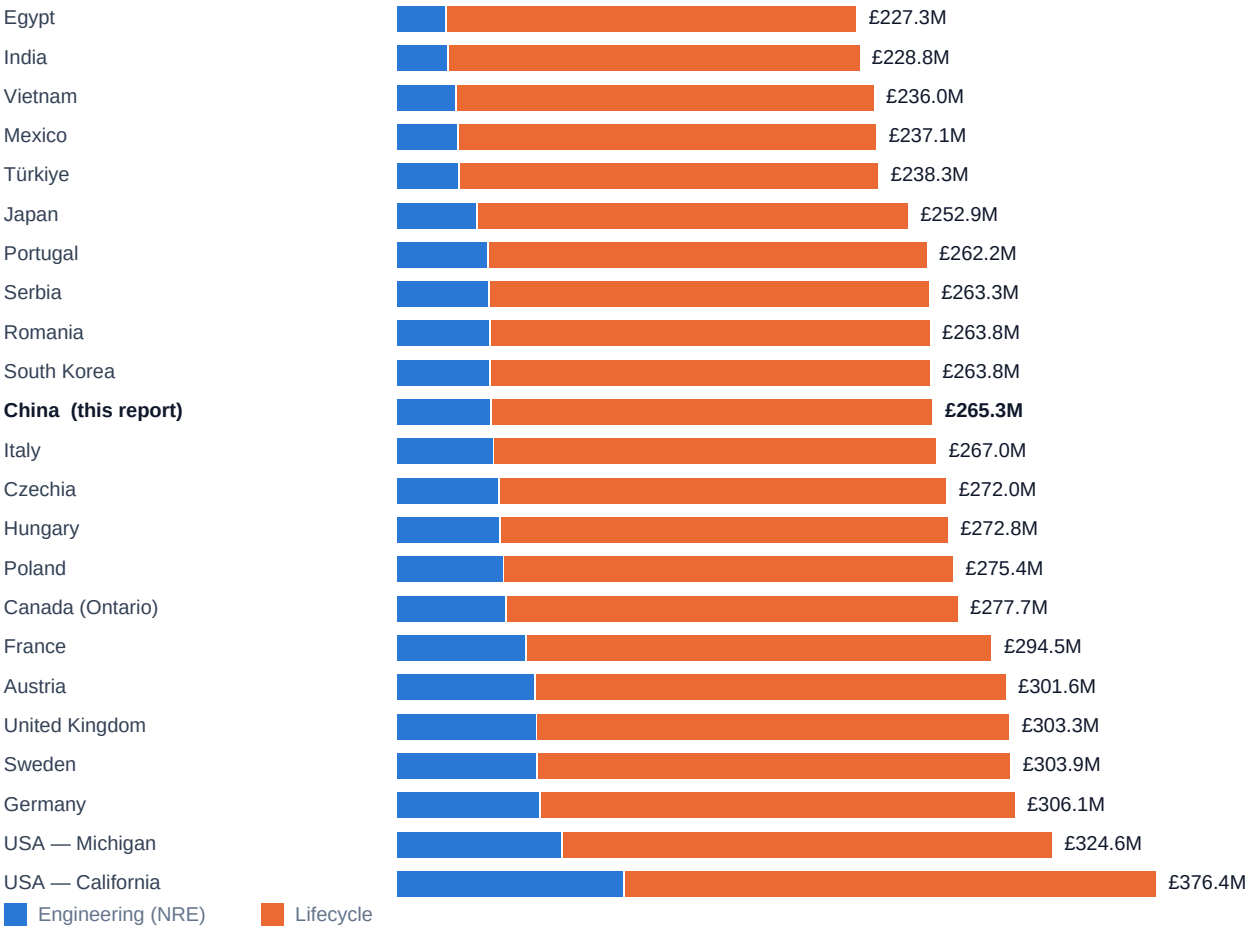
# 7 Spend by year

Nominal £ (no discounting). Engineering over 54 development months by phase share; maintenance in full for 3 years after SOP then a tail; royalties per vehicle built and for 3 subscription years; cloud for 8 years per connected vehicle, so service years run past the last build year.

| Year | Engineering | Maintenance | Licences | Royalties | Cloud | Total | Cumulative |
|---|---|---|---|---|---|---|---|
| Dev 1 | £10.4M | £0.0M | £0.0M | £0.0M | £0.0M | £10.4M | £10.4M |
| Dev 2 | £13.2M | £0.0M | £0.0M | £0.0M | £0.0M | £13.2M | £23.6M |
| Dev 3 | £11.3M | £0.0M | £0.0M | £0.0M | £0.0M | £11.3M | £34.8M |
| Dev 4 | £7.4M | £0.0M | £0.0M | £0.0M | £0.0M | £7.4M | £42.3M |
| Dev 5 | £4.0M | £0.0M | £0.0M | £0.0M | £0.0M | £4.0M | £46.3M |
| Year 1 | £0.0M | £3.2M | £1.8M | £9.9M | £2.2M | £17.0M | £63.2M |
| Year 2 | £0.0M | £3.2M | £1.8M | £12.2M | £4.3M | £21.4M | £84.7M |
| Year 3 | £0.0M | £3.2M | £1.8M | £14.5M | £6.5M | £25.9M | £110.6M |
| Year 4 | £0.0M | £1.6M | £1.8M | £14.5M | £8.6M | £26.4M | £137.0M |
| Year 5 | £0.0M | £1.6M | £1.8M | £14.5M | £10.7M | £28.6M | £165.6M |
| Year 6 | £0.0M | £1.6M | £1.8M | £14.5M | £12.9M | £30.7M | £196.3M |
| Service 1 | £0.0M | £0.4M | £0.0M | £4.6M | £12.9M | £17.9M | £214.2M |
| Service 2 | £0.0M | £0.4M | £0.0M | £2.3M | £12.9M | £15.6M | £229.8M |
| Service 3 | £0.0M | £0.4M | £0.0M | £0.0M | £10.7M | £11.1M | £241.0M |
| Service 4 | £0.0M | £0.4M | £0.0M | £0.0M | £8.6M | £9.0M | £250.0M |
| Service 5 | £0.0M | £0.4M | £0.0M | £0.0M | £6.5M | £6.8M | £256.8M |
| Service 6 | £0.0M | £0.4M | £0.0M | £0.0M | £4.3M | £4.7M | £261.5M |
| Service 7 | £0.0M | £0.4M | £0.0M | £0.0M | £2.2M | £2.5M | £264.1M |
| Service 8 | £0.0M | £0.4M | £0.0M | £0.0M | £0.0M | £0.4M | £264.5M |
| Service 9 | £0.0M | £0.4M | £0.0M | £0.0M | £0.0M | £0.4M | £264.9M |
| Service 10 | £0.0M | £0.4M | £0.0M | £0.0M | £0.0M | £0.4M | £265.3M |
| **Total** | **£46.3M** | **£18.4M** | **£10.6M** | **£86.8M** | **£103.2M** | **£265.3M** | |

# 8 Every engineering hub and every country

The same 52-module scope re-costed at each of the 23 hubs in the rate book, each at its own salaries, employer on-costs and working hours (no UK multiplier). Only engineering and its maintenance move with the hub.




| Hub | Rate confidence | Loaded £/PM | NRE | Lifecycle | Total | Per vehicle | v this report |
|---|---|---|---|---|---|---|---|
| Egypt | Low | £882 | £23.7M | £203.5M | £227.3M | £189 | -£38.0M |
| India | Low | £1,072 | £24.6M | £204.2M | £228.8M | £191 | -£36.5M |
| Vietnam | Low | £1,955 | £28.9M | £207.1M | £236.0M | £197 | -£29.3M |
| Mexico | Low | £2,101 | £29.6M | £207.6M | £237.1M | £198 | -£28.1M |
| Türkiye | Low | £2,240 | £30.2M | £208.0M | £238.3M | £199 | -£27.0M |
| Japan | Medium | £4,051 | £38.9M | £214.0M | £252.9M | £211 | -£12.3M |
| Portugal | Low | £5,198 | £44.5M | £217.8M | £262.2M | £219 | -£3.0M |
| Serbia | Medium | £5,333 | £45.1M | £218.2M | £263.3M | £219 | -£1.9M |
| Romania | Low | £5,390 | £45.4M | £218.4M | £263.8M | £220 | -£1.5M |
| South Korea | Medium | £5,396 | £45.4M | £218.4M | £263.8M | £220 | -£1.4M |
| **China** | Low | £5,573 | £46.3M | £219.0M | **£265.3M** | £221 | — |
| Italy | Low | £5,788 | £47.3M | £219.7M | £267.0M | £223 | +£1.7M |
| Czechia | Medium | £6,399 | £50.2M | £221.8M | £272.0M | £227 | +£6.7M |
| Hungary | Low | £6,500 | £50.7M | £222.1M | £272.8M | £227 | +£7.5M |

| Hub | Rate confidence | Loaded £/PM | NRE | Lifecycle | Total | Per vehicle | v this report |
|---|---|---|---|---|---|---|---|
| Poland | Low | £6,824 | £52.3M | £223.2M | £275.4M | £230 | +£10.1M |
| Canada (Ontario) | Medium | £7,108 | £53.6M | £224.1M | £277.7M | £231 | +£12.4M |
| France | Medium | £9,181 | £63.6M | £230.9M | £294.5M | £245 | +£29.2M |
| Austria | Low | £10,058 | £67.8M | £233.8M | £301.6M | £251 | +£36.4M |
| United Kingdom | Medium | £10,267 | £68.8M | £234.5M | £303.3M | £253 | +£38.0M |
| Sweden | Medium | £10,338 | £69.2M | £234.7M | £303.9M | £253 | +£38.6M |
| Germany | Medium | £10,605 | £70.4M | £235.6M | £306.1M | £255 | +£40.8M |
| USA — Michigan | Medium | £12,890 | £81.4M | £243.2M | £324.6M | £270 | +£59.3M |
| USA — California | Medium | £19,284 | £112.2M | £264.3M | £376.4M | £314 | +£111.1M |

## Every manufacturing country and its software hub (39)

Software is costed where it is engineered. A country with its own assessed hub uses it; any other uses the nearest assessed hub, for the reason given.

| Country | Software hub | Programme total at that hub | Why this hub |
|---|---|---|---|
| Austria | Austria | £301.6M | its own assessed hub |
| Belgium | Germany | £306.1M | Western Europe — Germany is the nearest assessed hub |
| Brazil | Mexico | £237.1M | Latin America — Mexico (Guadalajara) is the assessed hub |
| Bulgaria | Romania | £263.8M | South-East Europe — Romania is the nearest assessed hub |
| Canada | Canada (Ontario) | £277.7M | its own assessed hub |
| China | China | £265.3M | its own assessed hub |
| Czech Republic | Czechia | £272.0M | its own assessed hub |
| Egypt | Egypt | £227.3M | its own assessed hub |
| France | France | £294.5M | its own assessed hub |
| Germany | Germany | £306.1M | its own assessed hub |
| Hungary | Hungary | £272.8M | its own assessed hub |
| India | India | £228.8M | its own assessed hub |
| Indonesia | Vietnam | £236.0M | South-East Asia — Vietnam is the assessed hub |
| Italy | Italy | £267.0M | its own assessed hub |
| Japan | Japan | £252.9M | its own assessed hub |
| Lithuania | Poland | £275.4M | Baltic — Poland is the nearest assessed hub |
| Malaysia | Vietnam | £236.0M | South-East Asia — Vietnam is the assessed hub |
| Mexico | Mexico | £237.1M | its own assessed hub |
| Morocco | Egypt | £227.3M | North Africa — Egypt (Valeo Cairo) is the assessed North-African hub |
| Netherlands | Germany | £306.1M | Western Europe — Germany is the nearest assessed hub |
| Philippines | Vietnam | £236.0M | South-East Asia — Vietnam is the assessed hub |
| Poland | Poland | £275.4M | its own assessed hub |
| Portugal | Portugal | £262.2M | its own assessed hub |
| Romania | Romania | £263.8M | its own assessed hub |
| Serbia | Serbia | £263.3M | its own assessed hub |
| Singapore | Japan | £252.9M | high-cost Asian engineering — Japan is the nearest assessed hub |
| Slovakia | Czechia | £272.0M | Central Europe — Czechia is the nearest assessed hub |

| Country | Software hub | Programme total at that hub | Why this hub |
|---|---|---|---|
| Slovenia | Austria | £301.6M | Central Europe — Austria is the nearest assessed hub |
| South Africa | India | £228.8M | no assessed African hub south of Egypt — India, the largest offshore automotive software base |
| South Korea | South Korea | £263.8M | its own assessed hub |
| Spain | Portugal | £262.2M | Iberia — the research found no Spanish embedded-automotive salary data; Portugal (Bosch Braga) is the assessed Iberian hub |
| Sweden | Sweden | £303.9M | its own assessed hub |
| Taiwan | China | £265.3M | Greater China — China is the assessed hub |
| Thailand | Vietnam | £236.0M | South-East Asia — Vietnam is the assessed hub |
| Tunisia | Egypt | £227.3M | North Africa — Egypt (Valeo Cairo) is the assessed North-African hub |
| Turkey | Türkiye | £238.3M | its own assessed hub |
| United Kingdom | United Kingdom | £303.3M | its own assessed hub |
| United States | USA — Michigan | £324.6M | United States — Michigan (Detroit / Ann Arbor OEM software centres); choose California for Bay Area teams |
| Vietnam | Vietnam | £236.0M | its own assessed hub |

# 9 Risk and sensitivity

1,000 seeded trials; each costed line is drawn from a triangular range set by its evidence grade, with a 55 % programme-wide correlation. The headline adds each line's most likely value; because most money rests on unsourced lines whose risk runs upwards, it sits at about P16, not at the median.

| Measure | Programme total | Per vehicle |
|---|---|---|
| P10 | £254.3M | £212 |
| **P50 (median)** | **£301.3M** | **£251** |
| P90 | £358.0M | £298 |
| Mean | £304.2M | £254 |
| Headline (~ P16) | £265.3M | £221 |

## What moves the total — one driver at a time, everything else held

| Driver (low case vs high case) | Low | Base | High | Swing |
|---|---|---|---|---|
| Programme life (4 vs 9 years) | £195.2M | £265.3M | £370.4M | £175.2M |
| Engineering hub (Egypt vs USA — California) | £227.3M | £265.3M | £376.4M | £149.2M |
| Connected-service life (5 vs 15 years) | £226.6M | £265.3M | £355.6M | £129.0M |
| Effort catalogue (× 0.5 vs × 2) | £234.8M | £265.3M | £326.3M | £91.5M |
| Map subscription (1 vs 7 years) | £237.7M | £265.3M | £320.5M | £82.8M |
| Reuse level (Heavy vs Fresh) | £241.1M | £265.3M | £305.9M | £64.9M |
| Complexity (all modules: Medium vs Very High) | £249.6M | £265.3M | £272.1M | £22.5M |
| ASIL level (all modules: B vs D) | £264.7M | £265.3M | £283.3M | £18.6M |
| Company burden (20 % vs 60 %) | £258.8M | £265.3M | £271.7M | £12.9M |
| Security support after the last build (0 vs 15 years) | £261.3M | £265.3M | £267.3M | £6.0M |
| Production volume (300k vs 100k a year, per vehicle) | £200 | £221 | £284 | £84 |

# 10 Evidence and sources

How well each part of the total is supported, the basis of every stated assumption, and the rate build-up.

| Evidence grade | In the total | Share | Simulated range (low · high) |
|---|---|---|---|
| Your figures | £0.0M | 0.0 % | × 0.9 · × 1.15 |
| Calibrated to your actuals | £0.0M | 0.0 % | × 0.85 · × 1.25 |
| Published source | £22.7M | 8.6 % | × 0.85 · × 1.2 |
| Labelled estimate | £33.9M | 12.8 % | × 0.7 · × 1.6 |
| Unsourced catalogue | £208.7M | 78.7 % | × 0.6 · × 1.9 |

## Largest unsourced amounts — replace these first

| What | In the total |
|---|---|
| Cloud (per connected vehicle-year) | £103.2M |
| Royalties | £64.1M |
| Engineering effort (module catalogue) | £26.8M |
| IP licences (module catalogue) | £10.6M |
| Security support after the last build (share assumed) | £4.0M |

## Stated assumptions and their basis

| Assumption | Basis |
|---|---|
| **Connected service per vehicle** | Each connected vehicle served 8 years from build — the median of current OEM terms: GM OnStar Basics 8 yr (MY2025+), Tesla Standard Connectivity 8 yr (ordered after 20 Jul 2022), Toyota up to 10 yr, Mercedes-Benz mbrace Connect 5 yr (US); OEM pages, re-checked 14 Oct 2026. A 2G / 3G / 4G shutdown can end it sooner. |
| **Map-data subscription** | 3 years of map updates per vehicle — Nissan, Subaru, Hyundai, Honda (2021–24) include 3; some brands 5 or lifetime (OEM pages, read from search results). |
| **Cloud £ per connected vehicle-year** | Module cloud £ / yr ÷ reference fleet — an unsourced estimate (~ £10.8 per connected vehicle-year with every module in scope). No OEM publishes a back-end cost; the only list price, AWS IoT FleetWise (US$0.60 / 0.45 per vehicle-month, messages and storage extra; closed to new customers 30 Apr 2026), is US$5.4–7.2 per vehicle-year for data collection alone. |
| **Development window** | 54 months feasibility -> SOP: Volkswagen's stated development cycle (IAA 2023; target 36 months). GM "around four years" (2022); Chinese OEMs ~18–24 months. Set your programme's own. |
| **Development tools** | CostVision estimate £39,500 per engineer-seat-year — the module catalogue's tool £ at the default programme ÷ its engineer-years; no public seat price (Vector CANoe is quote-only). |
| **Maintenance profile** | Module % of development + verification + integration a year: in full for 3 years after SOP, then 50 % — rule of thumb (15–25 % a year, Edana; 21–26 % median years 1–3, 24 DACH SaaS products), not automotive-measured. |
| **Security support after the last build** | After the last build: each module's tail sustaining team × 25 % for 10 years — the duty is regulatory (UN R155: the post-production phase runs "until the end-of-life of all vehicles under the vehicle type"; R156: updates over the vehicles' lifetime); 10 years = Volkswagen / CARIAD's stated software support after end of production (2023, via the press); 25 % is a CostVision assumption — no published cost. |
| **Homologation** | Type approval ~ $1M (S&P Global: R155 on one legacy model "can easily exceed $1M"); ISO 26262 assessor £600k (ASIL-D) / £250k; organisation CSMS £1.5M + SUMS £0.8M ÷ vehicle types sharing them — CostVision estimates. |
| **Supplier margin (bought development)** | Each column re-costs the scope with that development source. A supplier's price = its effort factor ÷ (1 - 11.7 % margin), so buying is cheaper only if the supplier needs <= 88.3 % of the in-house effort (reuse, a cheaper hub mix). |
| **Effort catalogue (person-months per module)** | CostVision catalogue — no published automotive productivity source was found (the defence data in Clark & Madachy's cost-estimation manual is the next source to read). Calibrate on your own actuals or size modules in KSLOC. |
| **Reuse, complexity factors** | CostVision catalogue factors — no published source. |
| **ASIL factor** | Solcept engineering blog (× 1.4 to ASIL-B, × 1.3 B -> D); A and C interpolated — Low confidence. |

## Rate basis — what these figures were priced on

| | |
|---|---|
| **Rate book** | CostVision built-in v2.0.0 (country book 2026-10) (reviewed 2026-10) |
| **Engineering hub** | China — its own salaries, statutory employer on-costs and working hours (no UK multiplier) |
| **Hub rate (£ / person-month of 152 h, before overhead)** | mid £2,828 · senior £4,001 — CNY 254,483 mid / 360,000 senior gross a year + employer on-cost 25.7% at mid = 319,885 / 452,520; ÷ 1,936 productive h × 152 h = CNY 25,115 / 35,528 per person-month ÷ 8.88 CNY per £ = £2,828 / £4,001 |
| **Engineer rate used** | £3,473 — 55 % senior blend of the hub's mid and senior rates |
| **Development source** | Tier-1 supplier × 0.997 = efficiency 0.88 (unsourced) ÷ (1 - supplier margin 11.7 %) |
| **Engineering burden** | × 1.610 = employment +15 % × company +40 % × capital +0 % (on the hub rate) |
| **Loaded rate (£ / person-month)** | £5,573 |
| **Rate confidence** | Low — figures read from search results of the cited pages (Oct 2026); see the rate book |
| **Powertrain** | PHEV |
| **Platform annual volume** | not shared (this programme carries all of it) |
| **Annual volume × life** | 200,000 × 6 yr |
| **NRE recovery** | 6 yr |
| **Discount rate** | 0 % |
| **Development duration (tool licences)** | 54 months |
| **Cloud** | £10.75 per connected vehicle-year (the in-scope modules' catalogue £ / yr ÷ 440,000 reference fleet — unsourced estimate) × 100 % connected, each vehicle served 8 yr; fixed platform £0 / yr |
| **Map / per-vehicle-year royalties** | 3 yr per vehicle |
| **Development tools** | £39,500 per engineer-seat-year |
| **Maintenance profile** | full 3 yr after SOP, then 50 %; wage escalation 0 % / yr |
| **Security support after the last build** | 10 yr at 25 % of the tail team — After the last build: each module's tail sustaining team × 25 % for 10 years — the duty is regulatory (UN R155: the post-production phase runs "until the end-of-life of all vehicles under the vehicle type"; R156: updates over the vehicles' lifetime); 10 years = Volkswagen / CARIAD's stated software support after end of production (2023, via the press); 25 % is a CostVision assumption — no published cost. |
| **Homologation** | on — organisation CSMS / SUMS shared by 1 vehicle type(s) |
| **Nominal effort basis** | 52 catalogue · 0 COCOMO II size (2.94 × KSLOC^1.0997) · 0 custom PM |
| **Effort calibration** | none (model as published) |

# 11 Make or buy and the supplier check

Each domain's engineering re-costed with each development source at this hub. Each column re-costs the scope with that development source. A supplier's price = its effort factor ÷ (1 - 11.7 % margin), so buying is cheaper only if the supplier needs <= 88.3 % of the in-house effort (reuse, a cheaper hub mix).

| Domain | OEM in-house | Tier-1 supplier | Start-up |
|---|---|---|---|
| Powertrain & Battery | £12.0M | £12.0M | £10.6M |
| ADAS Level 2 & 2+ | £12.9M | £12.8M | £11.4M |
| Infotainment, Connectivity & UX | £5.9M | £5.9M | £5.2M |
| Vehicle Domain Controllers | £6.3M | £6.3M | £5.6M |
| Middleware & Platform | £2.4M | £2.4M | £2.1M |
| Cybersecurity (ISO 21434) | £1.4M | £1.4M | £1.2M |
| OTA & Cloud Backend | £1.8M | £1.8M | £1.6M |

A bought source pays only when its effort is below 88.3 % of the in-house effort. The efficiency factors behind it are unsourced — set your supplier's own.

## Supplier quote check

No supplier quote was entered on the Supplier check tab.

Negotiation target: a fair Tier-1 price for this scope is £46.3M of engineering at £5,573 per person-month (should-cost re-costed as a Tier-1 supplier: its effort factor ÷ (1 - 11.7 % margin)).

# 12 Scope, exclusions and limitations

## Catalogue modules not in this programme

| Module | Domain | Why not |
|---|---|---|
| Digital Key (UWB/BLE) & Secure Access | Infotainment, Connectivity & UX | not confirmed for this vehicle — excluded rather than assumed |
| 48 V Belt Starter-Generator & Battery Control | Powertrain & Battery | a 48 V mild-hybrid starter-generator; not this drivetrain |

## Vehicle software the catalogue has no module for (review judgement)

These functions are normally supplier ECUs whose software is priced in the hardware piece price. They are not in this total; confirm against the electrical bill of materials.

- Occupant restraint (airbag) control
- On-board AC charger and DC-DC converter control (the charging module covers DC fast-charge protocols)
- Electric power steering and electric parking brake base software
- Tyre-pressure monitoring
- Seat memory / massage and other comfort ECUs beyond climate and body control
- Trailer functions

## Limitations

- Accuracy has not been measured: no real programme cost has been logged against the model. The reconciliation in section 3 proves the arithmetic, not the inputs.
- Sourced figures (hub salaries, supplier margins, licence prices, OEM service terms) were read from search results of the cited pages, and each new default was re-checked by an independent search; the pages themselves could not be opened from the build environment.
- Published "programme software cost" figures for other vehicles are not printed: none could be verified.
- Vehicle volume and programme life are user inputs. The manufacturer publishes no model-level volume for this vehicle.
- The voice-engine royalty uses Cerence's published average price per unit as the reference price for an embedded voice licence; the vehicle's own voice-assistant terms are not public.

# A Appendix — China engineering rate build-up

Rate confidence: Low.

CNY 254,483 mid / 360,000 senior gross a year + employer on-cost 25.7% at mid = 319,885 / 452,520; ÷ 1,936 productive h × 152 h = CNY 25,115 / 35,528 per person-month ÷ 8.88 CNY per £ = £2,828 / £4,001

The book takes the median of the observations below: mid = median of 2 (238,966, 270,000) = CNY 254,483; senior = median of 1 (360,000) = CNY 360,000.

| Figure | Source | Page |
|---|---|---|
| Mid: CNY 238,966 | National Bureau of Statistics, 2024 (released 16 May 2025) — industry 'Information transmission, software and information technology services' (mean) | https://www.stats.gov.cn/sj/zxfb/202505/t20250516_1959826.html |
| Mid: CNY 22,500 | Shanghai advert, AUTOSAR CP, 5–10 y experience: 1.5–3 (zhaopin) (independent review) (midpoint of 15,000–30,000 a month × 12 (Shanghai AUTOSAR advert, 5–10 y)) | https://m-pre.zhaopin.com/jobs/CCL1318168350J40796784508.htm |
| Senior: CNY 30,000 | same advert (independent review) (advert range top × 12) | https://m-pre.zhaopin.com/jobs/CCL1318168350J40796784508.htm |
| Employer on-cost | pension 16% + medical incl. maternity 9% + unemployment 0.5% + work injury (lowest class) 0.2% — PwC Worldwide Tax Summaries China (Shanghai); housing provident fund (5–7 %) not sourced — excluded; base cap 300 % of city average not applied | https://taxsummaries.pwc.com/peoples-republic-of-china/individual/other-taxes |
| Working hours | (260 - 5 leave - 13 public holidays) × 40 h ÷ 5 — 5 days leave (1–10 years) — https://en.wikipedia.org/wiki/List_of_minimum_annual_leave_by_country; 13 public holidays not sourced in this research — CostVision reading of the national calendar / labour code, to verify | |